

SHORT QUESTIONS

Q.1 Differentiate between the deterministic model and the stochastic model.

Aspect	Deterministic Model	Stochastic Model
Definition	A model where the outcome is fully determined by the input variables and parameters, without any randomness involved.	A model that incorporates randomness or probability to account for uncertainty and variability in the outcomes.
Nature	Predictable and fixed: The same input will always produce the same output.	Probabilistic and variable: The same input can produce different outputs due to randomness.
Representation	Typically represented as equations without random terms.	Includes random variables or probabilistic components (e.g., noise).
Outcome	A single, precise outcome for a given set of inputs.	A range of possible outcomes, each associated with a probability.
Uncertainty	Assumes no uncertainty; all inputs and processes are known and fixed.	Models uncertainty by introducing probabilistic elements or distributions.
Complexity	Simpler and easier to interpret due to the absence of randomness.	More complex due to the need to account for randomness and variability.
Use Case	Used in systems where the relationship between inputs and outputs is exact and well-defined.	Used in systems where uncertainty, randomness, or variability plays a significant role.

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Examples	Newton's laws of motion.	Weather forecasting.
	Linear programming for optimization.	Stock price modeling.
	Population growth with fixed rates.	Economic demand prediction with random shocks.
Mathematical Form	$Y = f(X)$, where $f(X)$ is deterministic and fixed.	$Y = f(X) + u_i$, u_i is a random error term.
Advantages	Easy to compute and interpret.	Models uncertainty realistically.
	Requires less data.	Reflects real-world randomness.
Disadvantages	Unrealistic for complex or random systems.	Requires more advanced computation and statistical knowledge.
	It cannot account for variability or noise.	Results are less precise and harder to interpret.

Q.2

Discuss the statistical properties of the OLS estimator.

The Ordinary Least Squares (OLS) estimator is widely used in regression analysis because of its desirable statistical properties. These properties rely on the assumptions of the classical linear regression model. Below are the key statistical properties:

➤ **Linearity of the Estimator**

The OLS estimator is linear, meaning it is a linear combination of the dependent variable

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➤ **Unbiasedness**

The OLS estimator is **unbiased** if the assumptions of the classical linear regression model hold. This means the expected value of the OLS estimator equals the true value of the parameter:

$$E(\hat{\beta}) = \beta$$

➤ **Efficiency (Minimum Variance)**

The OLS estimator is **BLUE** (Best Linear Unbiased Estimator) under the Gauss-Markov theorem:

- It has the **minimum variance** among all linear and unbiased estimators.
- This property holds if the error term has a **constant** variance (homoscedasticity) and is uncorrelated across observations.

➤ **Consistency**

The OLS estimator is **consistent**, meaning that as the sample size (n) increases, the estimator converges to the true parameter value:

$$\lim_{n \rightarrow \infty} (\hat{\beta}) \rightarrow \beta$$

This property holds if the explanatory variables are not perfectly collinear and the error term has zero mean and finite variance.

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➤ **Normality (Asymptotic and Finite-Sample)**

- **Finite-Sample Normality:** If the error terms are normally distributed, the OLS estimator is normally distributed in finite samples:
- **Asymptotic Normality:** The OLS estimator is asymptotically normal even if the error terms are not normally distributed due to the Central Limit Theorem (for large samples).

➤ **Best Linear Unbiased Estimator (BLUE)**

Under the Gauss-Markov assumptions (linear relationship, no perfect multicollinearity, zero mean of errors, homoscedasticity, and no autocorrelation), the OLS estimator is BLUE:

- **Best:** Minimum variance.
- **Linear:** A linear function of the dependent variable.
- **Unbiased:** $E(\hat{\beta}) = \beta$

➤ **Sensitivity to Multicollinearity**

If multicollinearity (high correlation between independent variables) exists:

- The variance of the OLS estimator increases, leading to less precise estimates.
- However, the OLS estimator remains unbiased.

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➤ **Sensitivity to Heteroscedasticity**

If heteroscedasticity (non-constant variance of errors) is present:

- OLS estimators are still unbiased and consistent.
- However, they are no longer efficient (they lose the minimum variance property).

➤ **Sensitivity to Autocorrelation**

If autocorrelation (correlation between error terms) is present:

- OLS estimators are still unbiased and consistent.
- Efficiency is lost, and standard errors are biased, leading to unreliable hypothesis testing.

OLS results can be unreliable when assumptions like homoscedasticity, no multicollinearity, and no autocorrelation are violated, necessitating diagnostic tests and corrections.

Q.3

Discuss causes of Multicollinearity.

Multicollinearity occurs when two or more independent variables in a regression model are highly correlated, meaning they share significant linear relationships. This makes it challenging to determine the individual effect of each independent variable on the dependent variable. Below are the major causes of multicollinearity:

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➤ **Inherent Relationships Among Variables**

Certain independent variables may naturally be correlated because of underlying relationships. For example:

- **Economic indicators:** Variables like GDP, income, and consumption are often highly correlated because they are interdependent.
- **Demographic variables:** Age and years of work experience are often related.

➤ **Poor Model Specification**

Poorly designed models can lead to multicollinearity, such as:

- **Including redundant variables:** If the model includes two or more variables that represent the same phenomenon, such as "total income" and "monthly income."
- **Using variables derived from one another:** For example, including both "distance in kilometers" and "distance in miles" as predictors.

➤ **Dummy Variable Trap**

The dummy variable trap arises when all dummy variables for a categorical variable are included in the model without excluding one reference category. This leads to perfect multicollinearity because the dummies are linearly dependent. For example:

- If there are three categories (A, B, C), including all three dummy variables will cause multicollinearity.

➤ **Insufficient Variation in the Data**

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When the independent variables lack sufficient variation or are nearly constant across observations, it can lead to multicollinearity.

Examples include:

- Regressing on variables collected over a short time span where values do not vary much.
- Survey data with a homogeneous sample where all participants have similar characteristics.

➤ **Highly Correlated Variables Due to Measurement Issues**

Errors in data collection or measurement can lead to multicollinearity. Examples include:

- Variables recorded incorrectly or imprecisely (e.g., using proxies that are highly correlated with each other).
- Overlapping questions in surveys that yield similar responses (e.g., asking for income in two different ways).

➤ **Overfitting (Including Too Many Variables)**

When a regression model includes a large number of independent variables relative to the number of observations, some of the variables are likely to be highly correlated. This often happens in:

- Small datasets with many predictors.
- High-dimensional models where the variables overlap conceptually.

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➤ **Interaction Terms or Polynomial Terms**

Including interaction or polynomial terms in a model can lead to multicollinearity because these terms are derived from the original variables, making them correlated. For example:

- Interaction term: X_1X_2
- Polynomial term: X^2

➤ **Aggregation of Data**

When data are aggregated into larger units (e.g., regions or time periods), correlations between variables tend to increase. For example:

- Aggregating monthly data into yearly averages can inflate correlations.

➤ **Functional Dependencies**

When one independent variable is an exact or approximate mathematical function of another variable, it introduces multicollinearity. For example:

- Including both "height in meters" and "height in centimeters" in a model.

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➤ **Combining Time-Series and Cross-sectional Data**

In panel data or pooled datasets, multicollinearity can arise if time trends or regional differences cause some variables to move together systematically. For example:

- Inflation and GDP growth rates might be correlated across countries over time.

➤ **Multicollinearity Introduced by Standardization or Transformation**

Transformations like scaling or normalizing variables (e.g., standardizing variables into z-scores) can sometimes lead to multicollinearity if the transformation enhances correlations among variables.

Multicollinearity arises due to various reasons, including inherent relationships between variables, poor model design, measurement issues, and overfitting. While multicollinearity does not bias the OLS estimates, it inflates the standard errors, making it harder to determine the significance of individual predictors. Careful model design and diagnostic checks are essential to address and manage multicollinearity effectively.

Q.4

Differentiate between time series and cross-sectional data.

Time series data and cross-sectional data are two fundamental types of data structures in statistics and econometrics. They differ in their nature, purpose, and analysis methods. Below is a detailed differentiation:

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Aspect	Time Series Data	Cross-Sectional Data
Definition	Data is collected on a single entity over time.	Data is collected on multiple entities at a single point in time.
Structure	Observations are indexed by time (e.g., daily, monthly, yearly).	Observations are indexed by individual units (e.g., people, firms, regions).
Key Variable	Time is the primary variable of interest.	Entities (e.g., individuals, and firms) are the primary variable of interest.
Purpose	To study trends, patterns, and dynamics over time.	To study variation across different entities at a given time.
Example	Daily stock prices of a company over five years.	Income levels of households in a city in 2023.
	Monthly inflation rates of a country.	Test scores of students in a school.
Analysis Focus	Focuses on temporal relationships and changes over time.	Focuses on differences or relationships across entities.
Dependence in Observations	Observations are often autocorrelated, meaning past values influence current values.	Observations are generally assumed to be independent of each other.
Modeling Techniques	ARIMA (Autoregressive Integrated Moving Average)	OLS Regression
	Exponential Smoothing	Logistic Regression
	Time Trend Analysis	ANOVA

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	Graphical Representation	Line graphs are commonly used to show trends over time.	Bar charts, histograms, or scatterplots are used to compare entities.
	Seasonality and Trends	Seasonality (regular patterns over time) and trends are often present.	No seasonality or trend, as data represents a snapshot in time.
	Data Frequency	Data is collected at regular intervals (e.g., daily, monthly).	Data is collected at one point in time or in a short window.
	Example in Economics	Quarterly GDP of a country from 2000 to 2020.	Income distribution across households in a region for the year 2023.
		Monthly unemployment rates.	Literacy rates among states in a country.
	Challenges	Handling autocorrelation and non-stationarity.	Ensuring representativeness of the sample.
		Seasonality adjustments.	Avoiding multicollinearity.
➤ Time series data is suited for studying trends and dynamics over time for a single entity. ➤ Cross-sectional data is ideal for comparing multiple entities at a specific moment. Understanding their differences helps in choosing the right analytical methods and interpreting results accurately.			

Q.5	Ordinary Least Squares (OLS) is a statistical method used to estimate the parameters of a linear regression model. It is one of the most widely used techniques in econometrics and statistics due to its simplicity and desirable statistical properties under certain assumptions.
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OLS minimizes the sum of the squared differences between the observed values of the dependent variable (Y) and the values predicted by the regression model ($\hat{Y}$). The objective is to find the line (or hyperplane in multiple dimensions) that best fits the data.

Consider the regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + u_i$$

$Y \rightarrow$ Dependent variable

$(X_1, X_2, \dots, X_k) \rightarrow$ Independent variables

$\beta_0 \rightarrow$ Intercept

$(\beta_1, \beta_2, \beta_3, \dots, \beta_k)$ are the coefficients of the independent variables involved in the model, respectively?

$u_i \rightarrow$ is the error term (unexplained variation)

The OLS estimates (β_i) are obtained by minimizing:

$$\sum \hat{u}_i^2 = \sum (Y - \hat{Y})^2 = \sum (Y - \hat{\beta}_0 - \hat{\beta}_1 X_1 - \hat{\beta}_2 X_2 - \hat{\beta}_3 X_3 - \dots - \hat{\beta}_k X_k)^2$$

Steps in OLS estimation
Step.1) Specify the regression model
Step.2) Set the objective: minimize the residual
Residual = $Y - \hat{Y}$

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	Step.3) To obtain the estimates of regression coefficient, derive the function $\sum (\hat{u}_i)^2$ using partial derivative and equate it to zero	
	Step.4) Solve simultaneously to obtain: $\hat{\underline{\beta}} = (\underline{X}^T \underline{X})^{-1} (\underline{X}^T \underline{Y})$	

➤ The OLS estimators satisfy the properties of linearity unbiasedness, consistency, and efficiency.

➤ OLS is a foundational method in regression analysis, offering a straightforward approach to estimating relationships between variables. Its statistical properties and ease of implementation make it a key tool in various fields, but careful attention must be paid to its assumptions for valid results.

Q.6

Discuss various functional forms of regression function

The functional form of a regression model describes the relationship between the dependent variable (Y) and independent variable(s) (X). The choice of the functional form is crucial as it reflects the nature of the relationship and influences the interpretation of coefficients, fit of the model, and predictions.

Model name	Equation	Uses	Example
Linear	$Y = \beta_1 + \beta_2 X + u_i$	Applicable when the relationship is straightforward and additive.	Predicting salary (Y) based on years of experience (X).

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	Log-linear	$\ln Y = \beta_1 + \beta_2 \ln X + u_i$	To model relationships where proportional changes are more meaningful than absolute changes.	Modeling the relationship between income and consumption
	Log-lin	$\ln Y = \beta_1 + \beta_2 X + u_i$	To model relationships where Y grows or shrinks exponentially with X.	Population growth or economic growth models.
	Lin-log	$Y = \beta_1 + \beta_2 \ln X + u_i$	Useful when Y responds logarithmically to changes in X.	Modeling wages (Y) as a function of years of education (X).
	Reciprocal	$Y = \beta_1 + \beta_2 \left(\frac{1}{X} \right) + u_i$	To model asymptotic relationships.	The time taken to complete a task decreases with experience
	Log reciprocal	$\ln Y = \beta_1 - \beta_2 \left(\frac{1}{X} \right) + u_i$	Modeling relationships where diminishing returns are expected, such as income and utility.	Suppose we have Y (output) and X (capital investment) data.
	Polynomial regression models	$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_K X_K + u_i$	Suitable when the relationship between Y and X has curves or bends.	Modeling the effect of advertising spend (X) on sales (Y) where diminishing returns occur.

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		Logit/Probit Models	Logit model $P(Y = 1 / X) = \frac{1}{1 + e^{-(\beta_1 + \beta_2 X)}}$ Probit model $P(Y = 1 / X) = \Phi(\beta_1 + \beta_2 X)$	Binary outcome predictions, such as loan approval (yes/no).	To predict whether a borrower will default on a loan based on their economic and financial characteristics.
		Interaction Models	$Y = \beta_1 + \beta_2 X_1 + \beta_2 X_2 + \beta_3 (X_1 X_2) + u_i$	To capture joint effects of two variables on Y.	Effect of education (X1) and training (X2) on wages.
		Piecewise Regression (Spline Regression)	$Y = \begin{cases} \beta_1 + \beta_2 X + u_i, & X \leq c \\ \gamma_1 + \gamma_2 X + u_i, & X > c \end{cases}$	When the relationship between Y and X changes at specific thresholds.	House pricing models with separate effects for small vs. large properties.
		Quadratic Log Models	$Y = \beta_1 + \beta_2 \ln(X) + \beta_3 [\ln(X)]^2 + u_i$	To capture concave or convex relationships	Economic studies where income has diminishing effects on utility.

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Choosing the Functional Form:

- **Nature of Data:** Examine scatterplots or relationships between variables.
- **Model Fit:** Compare models using measures like R^2 , AIC, or BIC.
- **Interpretability:** Prefer simpler models if the fit is comparable.
- **Theory:** Follow theoretical knowledge to choose an appropriate form.

Each functional form captures different dynamics, and choosing the right one enhances model accuracy and insights.

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Long Questions

Q.2

The estimated model pertaining to child mortality (CM) on per capita GNP (PGNP) and the female literacy rate (FLR) for a sample of 64 countries is reported below along respective standard errors in parenthesis.

$$\hat{CM} = 263.6416 - 0.0056PGNP - 2.2316FLR$$

$$se = (11.5932), (0.0019), (0.2099)$$

- Is coefficient of PGNP statistically significant?
- Is coefficient of FLR statistically significant?
- Are both coefficients statistically significant?

Solution:

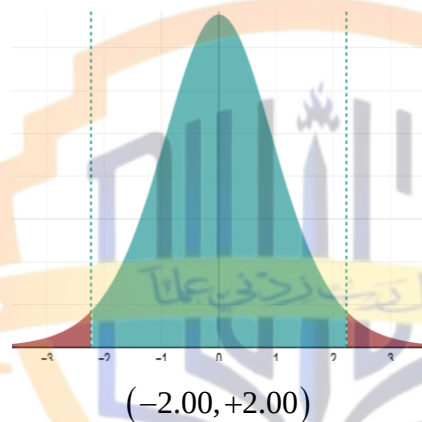
Is the coefficient of PGNP statistically significant?	Is the coefficient of FLR statistically significant?
Step.1) State the hypotheses	Step.1) State the hypotheses
Step.2) Level of significance $\alpha = 0.05$	Step.2) Level of significance $\alpha = 0.05$
Step.3) Test statistic $t = \frac{\hat{\beta}_1 - \beta_1}{S.E(\hat{\beta}_1)}$	Step.3) Test statistic $t = \frac{\hat{\beta}_2 - \beta_2}{S.E(\hat{\beta}_2)}$
Step.4) Calculation $t = \frac{\hat{\beta}_1 - \beta_1}{S.E(\hat{\beta}_1)} = \frac{-0.0056 - 0}{0.0019} = -2.95$	Step.4) Calculation $t = \frac{\hat{\beta}_2 - \beta_2}{S.E(\hat{\beta}_2)} = \frac{-2.2316}{0.2099} = -10.63$
Step.5) Critical Region	Step.5) Critical Region

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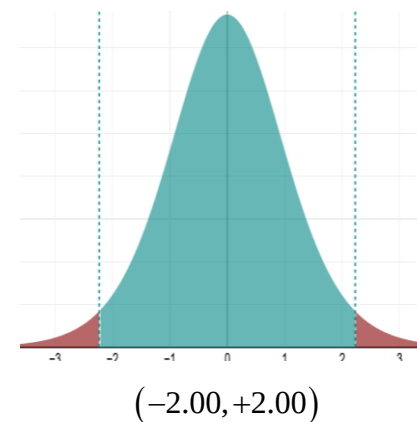
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at d.f = n-2 = 62

**Step.6) Conclusion**

- **Statistical Significance:** The t-statistic ($t = -2.95$) exceeds the critical value: $t_{\frac{\alpha}{2}(v)} = \pm 2.00$, indicating that PGNP is statistically significant at the 5% level.
- **Interpretation:** For every 1-unit increase in PGNP, child mortality is expected to increase by 0.0056 units, holding the female literacy rate constant.

at d.f = n-2 = 62

**Step.6) Conclusion**

- Statistical Significance:** The t-statistic ($t = -10.63$) far exceeds the critical value: $t_{\frac{\alpha}{2}(v)} = \pm 2.00$, indicating that FLR is statistically significant at the 5% level.
- Interpretation:** For every 1-unit increase in FLR, child mortality is expected to decrease by 2.2316 units, holding PGNP constant.

Both PGNP and FLR are statistically significant predictors of child mortality. However, the positive coefficient for PGNP raises questions about the underlying dynamics.

- The strong negative relationship between FLR and child mortality suggests that investing in female education could be a highly effective way to reduce child mortality.

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| | ➤ The unexpected positive association between PGNP and child mortality warrants further investigation, possibly by exploring non-linear relationships, interactions, or confounding factors. |
|--|--|

Q.3

Define Dummy variable. How can avoid dummy trap? What are the uses of dummy variables in regression analysis?

A **dummy variable** is a numerical variable used in regression analysis to represent categorical data. It takes the value of 0 or 1 to indicate the absence or presence of a certain condition or category. Dummy variables allow the inclusion of categorical data in regression models by converting qualitative information into a quantitative form.

Example:

For a categorical variable like "Gender" with two categories, "Male" and "Female":

- Dummy variable $D_1 = 1$ for "Male", $D_1 = 0$ for "Female".

For a variable like "Color" with three categories (Red, Blue, Green), we create two dummy variables:

- $D_1 = 1$ if Red, 0 otherwise.
- $D_2 = 1$ if Blue, 0 otherwise. (Here, Green is implicitly represented when $D_1 = 0$ and $D_2 = 0$)

How to Avoid the Dummy Variable Trap?

The **dummy variable trap** occurs when the inclusion of all dummy variables for a categorical variable introduces **multicollinearity** in the regression model. This happens because the dummy variables are linearly dependent, violating the assumption of no perfect multicollinearity in regression.

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To avoid the dummy variable trap:

1. If a categorical variable has k categories, create $k-1$ dummy variables.
 - For example, for a variable with three categories (Red, Blue, Green), create only two dummy variables.
2. Leave one category as the **reference category**. The regression intercept will capture the effect of the reference category.

Uses of Dummy Variables in Regression Analysis**➤ Inclusion of Categorical Variables:**

- Dummy variables allow categorical data (e.g., gender, region, education level) to be included in regression models, enabling analysis of their effects on the dependent variable.

➤ Interaction Effects:

- Dummy variables can be interacted with other variables (e.g., continuous variables) to study how the effect of one variable changes across categories.

➤ Group Comparisons:

- They help compare the mean differences in the dependent variable across different categories. For example, assessing whether salaries differ by gender.

➤ Policy Analysis:

- Dummy variables are used to evaluate the impact of policy changes by including a dummy for periods before and after the policy implementation.

➤ Seasonal Adjustments:

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- Dummy variables can represent seasons (e.g., winter, spring) or time periods (e.g., quarters in a year) to capture seasonal effects in time-series data.

➤ **Segmentation Analysis:**

- They can help segment data based on categories and identify how different segments behave in relation to the dependent variable

Q.4

Obtain the regression line (Y and X), interpret the results, calculate the standard error of the coefficient, and comment on the statistical significance of the X-variable.

Y	2.75	2.15	4.40	5.50	3.20	4.30	2.31	4.30	3.70
X	29.50	26.30	32.20	36.50	27.20	27.50	28.30	30.30	28.50

Solution:

Y	X	$(Y - \bar{Y})$	$(X - \bar{X})$	$(Y - \bar{Y})^2$	$(X - \bar{X})^2$	$(X - \bar{X})(Y - \bar{Y})$	$\hat{Y} = \hat{\beta}_1 + \hat{\beta}_2 X = -4.487 + 0.274X$
2.75	29.50	-0.87	-0.09	0.7569	0.0081	0.0783	3.5960
2.15	26.30	-1.47	-3.29	2.1609	10.8241	4.8363	2.7192
4.40	32.20	0.78	2.61	0.6084	6.8121	2.0358	4.3358
5.50	36.50	1.88	6.91	3.5344	47.7481	12.9908	5.5140
3.20	27.20	-0.42	-2.39	0.1764	5.7121	1.0038	2.9658
4.30	27.50	0.68	-2.09	0.4624	4.3681	-1.4212	3.0480
2.31	28.30	-1.31	-1.29	1.7161	1.6641	1.6899	3.2672
4.30	30.30	0.68	0.71	0.4624	0.5041	0.4828	3.8152
3.70	28.50	0.08	-1.09	0.0064	1.1881	-0.0872	3.322
32.61	266.30	0.03	-0.01	9.88	78.83	21.61	

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$$Y = \beta_1 + \beta_2 X + u_i$$

$$\hat{Y} = \hat{\beta}_1 + \hat{\beta}_2 X$$

$$\hat{\beta}_2 = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2} = \frac{21.61}{78.83} = 0.274$$

$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X} = \left(\frac{32.61}{9} \right) - \left(0.2741 \times \frac{266.30}{9} \right) = 3.6233 - (0.2741 \times 29.5888) = 3.6233 - 8.1103 = -4.487$$

$$\hat{Y} = \hat{\beta}_1 + \hat{\beta}_2 X = -4.487 + 0.274X$$

$\hat{Y}$	Y	$(Y - \hat{Y})$	$(Y - \hat{Y})^2$
3.596	2.75	-0.85	0.715716
2.7192	2.15	-0.57	0.323989
4.3358	4.40	0.06	0.004122
5.514	5.50	-0.01	0.000196
2.9658	3.20	0.23	0.054850
3.048	4.30	1.25	1.567504
3.2672	2.31	-0.96	0.916232
3.8152	4.30	0.48	0.235031
3.322	3.70	0.38	0.142884
		0.03	3.960523

$$R^2 = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2} = 1 - \frac{3.960523}{9.88} = 1 - 0.401 = (0.599)100\% = 59.9\%$$

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$$\therefore \sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum (Y - \hat{Y})^2$$

$$9.88 = \sum (\hat{Y} - \bar{Y})^2 + 3.960523$$

$$9.88 - 3.960523 = \sum (\hat{Y} - \bar{Y})^2$$

$$5.919477 = \sum (\hat{Y} - \bar{Y})^2$$

$$F = \frac{\left(\frac{\sum (\hat{Y} - \bar{Y})^2}{1} \right)}{\left(\frac{\sum (Y - \hat{Y})^2}{N - 2} \right)} = \frac{5.919477}{\frac{3.960523}{7}} = \frac{5.919477}{1} \times \frac{7}{3.960523} = 10.46$$

- **Intercept (-4.487):** When $X=0$, the predicted value of Y would be -4.4879 . However, since X does not realistically reach 0 in this dataset, the intercept primarily serves as a mathematical adjustment.
- **Slope (0.274):** On average, for every 1-unit increase in X , Y is expected to increase by 0.2741 units.
- **R-squared = 0.599:** About 59.9% of the variance in Y is explained by X .
- $\therefore F_{\alpha(1,7)} = 5.59, F = 10.46$, suggests that the overall model is statistically significant. The independent variable X has a statistically significant positive effect on Y , and the regression model provides a moderately good fit for the data.

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$$S.E(\hat{\beta}_2) = \frac{S_{Y.X}}{\sqrt{\sum (X - \bar{X})^2}} = \frac{\sqrt{\frac{\sum (Y - \hat{Y})^2}{n-2}}}{\sqrt{\sum (X - \bar{X})^2}} = \frac{\sqrt{\frac{3.960523}{7}}}{\sqrt{78.83}} = 0.085$$

$$t = \frac{\hat{\beta}_2 - \beta_2}{S.E(\hat{\beta}_2)} = \frac{0.2741}{0.085} = 3.22$$

$$t_{\frac{\alpha}{2}(n-2)} = t_{0.025(7)} = \pm 2.3646$$

The calculated value of t lies in the critical region. This indicates that X has a meaningful relationship with Y.

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